

the measurement electronics. So it is in these non-electronic areas where most effort must go in order to improve the overall reliability of the device.

Freescale offers comprehensive solutions that help original equipment manufacturers quickly develop ultra-low-power, portable and wireless medical products. Its activity monitor reference design allows recorded data to be interpreted and summarised for improved health performance, enabling consumers to better manage their fitness activities.

Cost. The same technology trends that make MCUs and other electronic components smaller also make them less expensive. Each new generation of semiconductor-fabrication technology allows components to be made smarter at a lower cost. Perhaps, more significant is the fact that as portable medical-electronic devices become more available and commonplace, these enable broader distribution of healthcare to more patients without requiring more doctors. This has a large impact on the overall cost of healthcare.

Dramatic reductions in cost have been achieved by using application-specific semiconductor solutions, as well as low-cost components from high-volume collateral markets such as mobile phones and automobiles.

Latest developments

A very interesting area for the development of new portable medical-electronic devices is drug delivery. We have seen devices that help people take medicines using a variety of methods. These devices contain MCUs and a variety of sensors and actuators to improve the efficiency and accuracy of medicine delivery.

A fascinating example is the electronic drug patch. This patch is affixed to the patient's skin using some adhesive like a bandage. The material of the patch is covered with a specially formulated drug, which is pushed through the skin into the blood by a controlled electric current. The process is completely painless,

MAJOR CONTRIBUTORS TO THIS REPORT



Sapna Mongia
South Asia Manager, Sensors
and High Performance
Analog Division, Analog,
Power and MEMS Group,
STMicroelectronics



Steve Kennelly
Senior Manager,
Medical Products Group,
Microchip Technology Inc.



Tom O'Dwyer
Director of Technology
for Healthcare, Analog
Devices Inc. (ADI)



Vivek Tyagi
Country Sales Director,
Freescale Semiconductor
India

and can be self-administered by the patient at home.

Other devices can deliver drugs by injection, inhalation, swallowing or drops in the eyes. In each case, the use of electronics results in more accurate dosing and an automated record to be kept of each dose, resulting in better treatment for the patient.

In addition, many older portable medical-electronic devices are being improved through the addition of friendlier user interfaces. Non-moving touch buttons, proximity detectors and touch screens, which are similar to interfaces on the latest smartphones, are making medical devices easier to use while reducing the possibilities for user error. As these advantages become more broadly known, medical device designers are paying more attention to this type of human-factors engineering.

Freescale Semiconductor recently introduced a home health hub (HHH) reference platform to help medical equipment manufacturers quickly and easily create remote-access devices that can collect, connect and securely share health data for improved healthcare management. The reference platform is based on Freescale's i.MX28 applications processor and ZigBee and sub-1GHz transceivers. It enables secure Wi-Fi and Ethernet connectivity to remote devices with displays, such as tablets, smartphones or PCs with medical-specific remote user interface options. The platform can also provide wired and wireless connectivity to end-point healthcare devices, such as

blood pressure monitors, blood glucometers, weighing scales and pulse oximeters.

Coming soon...

Going forward, it is expected that body-worn sensors will become a very large market. Technical problems remain to be overcome in the sensors required, the algorithms which interpret them and the wireless connectivity required to get the information displayed on an aggregator device such as a mobile phone. These challenges are actively being addressed and we are starting to see emergence of the first devices.

Monica Healthcare, a UK-based company, has launched a disposable adhesive patch device for monitoring pregnant mothers close to delivery. The patch can sense mother and fetus heartbeats as well as abdominal contractions. We can expect to see many more patch devices entering the market.

Also, the introduction of cloud has added a new dimension to the healthcare scenario. 3D printing has been around for a while but now it's being applied to medicine.

At the recently concluded *Consumer Electronics Show* in Las Vegas, we saw introduction of a variety of health devices such as Wi-Fi smart scale and sensor patch that if worn for seven days discreetly monitors the activity of the user's body. This signifies a growing interest in portable medical devices in the consumer segment. ●

The author is an executive editor at EYF